

Aunque Java se utiliza normalmente para desarrollar aplicaciones a medida, tenemos varios CMS contruidos con Java.

En esta tarea vamos a hacer un despliegue sencillo de una aplicación Java: OpenCMS. El problema que tenemos que este CMS no es compatible con tomcat10, pero no hay problemas lo vamos a instalar sobre un contenedor docker, un Tomcat 9:

Tener Docker y Docker compose instalado :

- 1ra parte : Preparar el Docker para desplegar el servidor TOMCAT

```
root@serveraweblamp:/home/jens# apt install docker
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
docker ya está en su versión más reciente (1.5-2).
0 actualizados, 0 nuevos se instalarán, 0 para eliminar y 0 no actualizados.
root@serveraweblamp:/home/jens#

root@serveraweblamp:/home/jens# sudo apt install docker-compose-plugin -y
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
docker-compose-plugin ya está en su versión más reciente (2.40.3-1ubuntu.22.04~jammy)
.
0 actualizados, 0 nuevos se instalarán, 0 para eliminar y 0 no actualizados.
root@serveraweblamp:/home/jens#
```

- 2da parte : Descarga el instalador de OpenCMS (descargar el .war y subirlo al servidor donde estará el docker)

```
root@serveraweblamp:/home/jens/opencms-docker# wget https://github.com/alkacon/opencms-core/releases/download/build_20_0_0/opencms-20.0-distribution.zip -O opencms-distribution.zip
--2025-11-27 14:12:45-- https://github.com/alkacon/opencms-core/releases/download/build_20_0_0/opencms-20.0-distribution.zip
Resolving github.com (github.com)... 140.82.121.3
Connecting to github.com (github.com)|140.82.121.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/1965842/fb01d8e5-05fd-4673-bb67-219fbb424940?sp=r&sv=2018-11-09&sr=b&spr=https&se=2025-11-27T14%3A47%3A22Z&rscd=attachment%3B+filename%3Dopencms-20.0-distribution.zip&rsc
root@serveraweblamp:/home/jens/opencms-docker# unzip opencms-distribution.zip
Archive: opencms-distribution.zip
  inflating: opencms.war
  inflating: history.txt
  inflating: LICENSE
  inflating: README.md
  inflating: INSTALL.md
root@serveraweblamp:/home/jens/opencms-docker#

ment%3B+2011+filename%3Dopencms-20.0-distribution.zip&response-content-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.111.133, 185.199.108.133, 185.199.109.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 236407720 (225M) [application/octet-stream]
Saving to: 'opencms-distribution.zip'

opencms-distribution. 100%[=====>] 225,46M  37,9MB/s   in 5,7s

2025-11-27 14:12:51 (39,8 MB/s) - 'opencms-distribution.zip' saved [236407720/236407720]

root@serveraweblamp:/home/jens/opencms-docker#
```

```

root@serveraweblamp:/home/jens/opencms-docker# ls -la tomcat/webapps/
total 230752
drwxr-xr-x 2 root root      4096 nov 27 14:18 .
drwxr-xr-x 4 root root      4096 nov 27 14:10 ..
-rw-r--r-- 1 root root 236276102 oct 13 13:07 opencms.war
root@serveraweblamp:/home/jens/opencms-docker#

```

- Hacer la estructura de carpetas

```

root@serveraweblamp:/home/jens/opencms-docker# ls -l
total 231236
-rw-r--r-- 1 root root      803 nov 27 14:24 docker-compose.yml
-rw-r--r-- 1 root root    323358 oct 13 12:29 history.txt
-rw-r--r-- 1 root root      5521 oct 13 12:29 INSTALL.md
-rw-r--r-- 1 root root     24425 oct 13 12:29 LICENSE
-rw-r--r-- 1 root root 236407720 oct 13 11:17 opencms-distribution.zip
-rw-r--r-- 1 root root      5640 oct 13 12:29 README.md
drwxr-xr-x 4 root root      4096 nov 27 14:10 tomcat
root@serveraweblamp:/home/jens/opencms-docker# cd tomcat/
root@serveraweblamp:/home/jens/opencms-docker/tomcat# ls -l
total 8
drwxr-xr-x 2 root root 4096 nov 27 14:10 logs
drwxr-xr-x 2 root root 4096 nov 27 14:18 webapps
root@serveraweblamp:/home/jens/opencms-docker/tomcat#

```

- Puedes seguir la [documentación](#) de OpenCMS.
- Tienes que descargar el fichero war para realizar el despliegue.
- Necesitas una base de datos MariaDB para que el CMS funcione.
 - Lo más fácil es crear un docker compose que te ayude a desplegar los dos contenedores.

Cuando este desplegado, accede a la aplicación y realiza alguna modificación para que aparezca tu nombre en la página principal.

Los pasos para esta práctica son :

- 3ra parte : Crear el archivo docker-compose.yml

```
root@serveraweb1amp: /home/jens/opencms-docker
GNU nano 6.2 docker-compose.yml *
version: '3.8'

services:
  mariadb:
    image: mariadb:10.11
    container_name: opencms-db
    restart: unless-stopped
    environment:
      MYSQL_ROOT_PASSWORD: rootpassword
      MYSQL_DATABASE: opencms
      MYSQL_USER: opencms
      MYSQL_PASSWORD: opencmspassword
    volumes:
      - db_data:/var/lib/mysql
    networks:
      - opencms-network

  tomcat:
    image: tomcat:9.0-jdk11-openjdk
    container_name: opencms-tomcat
    restart: unless-stopped
    ports:
      - "8082:8080"
    environment:
      CATALINA_OPTS: "-Xmx1024m -Xms512m"
    volumes:
      - ./tomcat/webapps:/usr/local/tomcat/webapps
      - ./tomcat/logs:/usr/local/tomcat/logs
    depends_on:
      - mariadb
    networks:
      - opencms-network

volumes:
  db_data:

networks:
  opencms-network:
    driver: bridge
```

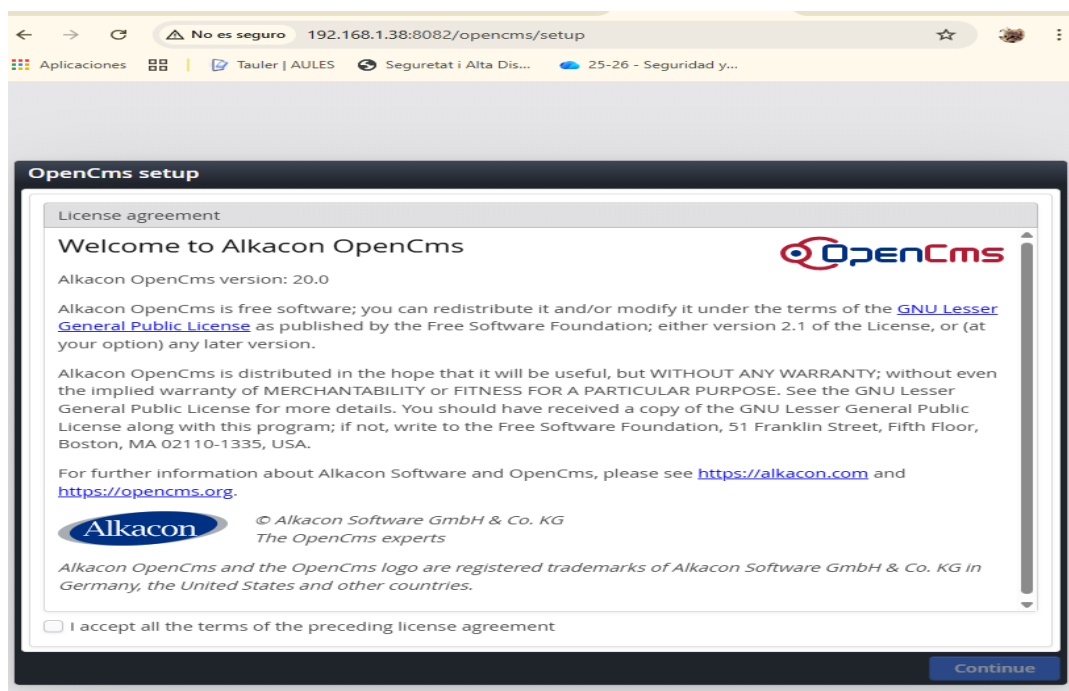
- Importante : Si utilizais una MV que ya tenga un docker con alguna aplicación ejecutándose en el puerto 8080, para esta deberéis cambiarla al 8081.

- 4ta parte : Levantar el stack (obvio)
- 5ta parte : Configurar el Open CMS

```
root@serveraweb1amp:/home/jens/opencms-docker# docker compose up -d
WARN[0000] /home/jens/opencms-docker/docker-compose.yml: the attribute `version` is ob
solete, it will be ignored, please remove it to avoid potential confusion
[+] Running 13/20
  ⚙️ mariadb [#####] 75.28MB / 106.3MB Pulling                               9.1s
  ✓ cffdfa9765de Download complete                                           0.4s
  ✓ 2540807a6100 Download complete                                           1.0s
  ✓ 892acd9bdcf5 Download complete                                           0.9s
  ⚙️ 0aa9032ee110 Downloading 40.49MB/71.13MB                               7.0s
  ✓ 96d7347ba317 Download complete                                           1.0s
  ✓ 9629f042e43d Download complete                                           0.9s
  ⚙️ 7e49dc6156b0 Extracting 2 s                                             7.0s
  ✓ 62c1a43ca7d9 Download complete                                           2.7s
  ⚙️ tomcat [#####] 160.9MB / 346.3MB Pulling                               9.1s
  ✓ fd3dc2f7be02 Download complete                                           1.1s
  ⚙️ 001c52e26ad5 Downloading 40.89MB/55MB                                   7.2s
  ✓ d9d4b9b6e964 Download complete                                           2.7s
  ✓ 2068746827ec Download complete                                           3.7s
  ⚙️ 9daef329d350 Downloading 46.14MB/54.58MB                               7.2s
  ✓ d85151f15b66 Download complete                                           2.5s
  ✓ 66223a710990 Download complete                                           0.6s
  ⚙️ db38d58ec8ab Downloading 39.85MB/202.1MB                               7.2s
  ✓ 39665df2e7f1 Download complete                                           1.0s
  ✓ 829da547da15 Download complete                                           3.8s
```

```
root@serveraweb1amp:/home/jens/opencms-docker# docker compose up -d
WARN[0000] /home/jens/opencms-docker/docker-compose.yml: the attribute `version` is obsolete, it will be ignored,
please remove it to avoid potential confusion
[+] Running 4/4
  ✓ Network opencms-docker_opencms-network Created                        0.1s
  ✓ Volume opencms-docker_db_data Created                                0.0s
  ✓ Container opencms-db Started                                         0.6s
  ✓ Container opencms-tomcat Started                                     0.8s
root@serveraweb1amp:/home/jens/opencms-docker#
```

- http://IP_DEL_SERVIDOR:8081/opencms/setup



OpenCms setup

Component Tests

Folder Permissions	passed
JDK Version	11.0.16
Temp Directory	/usr/local/tomcat/temp
Operating System	Linux 5.15.0-161-generic
Servlet Container	Apache Tomcat/9.0.65
Image Processing	passed
Unpacked WAR File	passed
Xerces Version	Xerces-J 2.12.2
XML API	passed

Status

Your system uses components which have been tested to work properly with Alkacon OpenCms.

[Back](#) [Continue](#)

OpenCms setup

Database product

MySQL/MariaDB >= 5.5 ▼

Setup connection

User

Password

OpenCms connection

User

Password

Database configuration

Connection string

Database name

☒ Create database and tables

☐ Drop database first if it already exists

[Back](#) [Continue](#)

Entregable típico para la práctica

1. Capturas de los pasos de la instalación : comandos, portainer como que está todo instalado... etc..
2. Captura donde se vea OpenCMS funcionando.
3. Captura del Workplace.
3. Alguna modificación de algún texto en la página principal (añadir su nombre).

OpenCms setup

Module groups available for installation

- ☒ OpenCms Workplace - Always required for a full OpenCms installation.
- ☒ Alkacon OpenCms Demo Template - The OpenCms demonstration template. Install this for a good demonstration of the OpenCms functionality.
- ☐ Legacy - Support for legacy XML pages or workplace tools written for OpenCms 9 or older. Install this only if you know you need it.

OpenCms setup

Server settings

Server ethernet address 5A:E4:61:BB:B2:DE

Server URL http://192.168.1.38:8082

Server ID OpenCmsServer

Back

Continue

OpenCms setup - Importing modules

OpenCms WEB-INF path: "/usr/local/tomcat/webapps/opencms/WEB-INF/".

OpenCms property file: "/usr/local/tomcat/webapps/opencms/WEB-INF/config/opencms.properties".

(26 / 141) Publishing file /system/modules/org.opencms.base/schemas/notification-user-password.xsd ... o.k.
(27 / 141) Publishing file /system/modules/org.opencms.base/schemas/notification.xsd ... o.k.
(28 / 141) Publishing file /system/workplace/commons/about.jsp ... o.k.
(29 / 141) Publishing file /system/workplace/commons/direct_publish.jsp ... o.k.
(30 / 141) Publishing file /system/workplace/commons/download-aliases.jsp ... o.k.
(31 / 141) Publishing file /system/workplace/commons/editor-backlink.html ... o.k.
(32 / 141) Publishing file /system/workplace/commons/gallery.jsp ... o.k.
(33 / 141) Publishing file /system/workplace/commons/import-aliases.jsp ... o.k.
(34 / 141) Publishing file /system/workplace/commons/open_ade_gallery.jsp ... o.k.
(35 / 141) Publishing file /system/workplace/commons/postupload.jsp ... o.k.
(36 / 141) Publishing file /system/workplace/commons/postupload_basicproperties.jsp ... o.k.
(37 / 141) Publishing file /system/workplace/commons/postupload_useconfig.jsp ... o.k.
(38 / 141) Publishing file /system/workplace/commons/properties.jsp ...

```
Indexing statistics: indexed files: 7,125, returned threads: 7,125, abandoned threads: 0, duration: 00:03:47
Admin@Offline>
Admin@Offline># Restore index modes
Admin@Offline>setIndexRebuildMode 'Solr Online' 'auto'
Admin@Offline>setIndexRebuildMode 'Solr Offline' 'offline'
Admin@Offline>
Admin@Offline>updateConfiguration
Admin@Offline>
Admin@Offline>exit
```

The setup is finished!
The OpenCms system used for the setup will now shut down.

Continue

OpenCms setup

Final notes

Congratulations - you have installed OpenCms



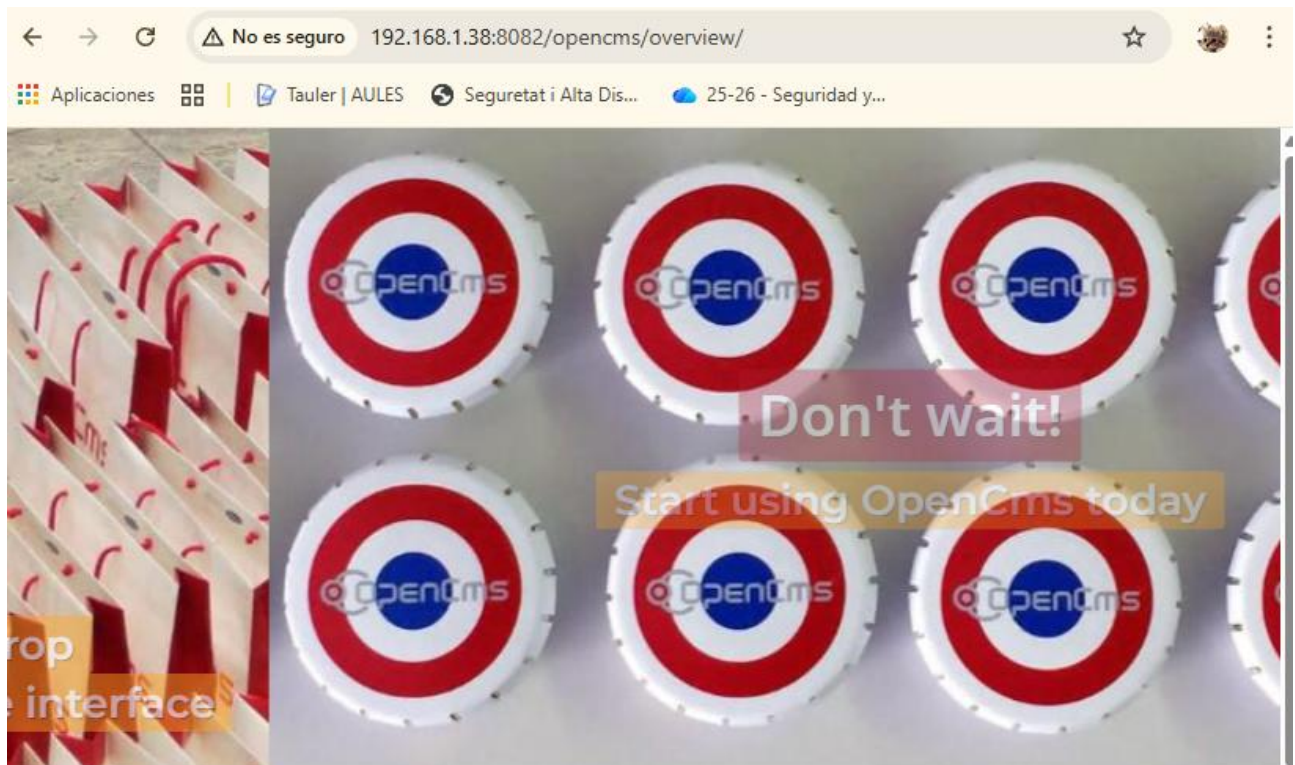
Alkacon OpenCms version: 20.0

Your next step is to explore the OpenCms workplace. The OpenCms workplace is where you can edit existing pages, create new pages, manage users etc. The workplace has been tested with recent versions of **Chrome**, **Firefox**, **Edge** and **Safari** browsers. Internet explorer 11 also works but is not recommended.

Please note: The browser requirements apply *only* when accessing the OpenCms workplace. For websites built with Alkacon OpenCms, there are no such restrictions.

For more information on the workplace and on developing in OpenCms, see <https://documentation.opencms.org>.

Continue



OpenCms Beginners Tutorial



Click here for a guided tour that explains in 6 simple steps how to edit content, create pages and work with OpenCms in general.

Documentation



The central documentation source on the web, including a detailed